

Abstract**Stochastic Model for Determining State of Charge and Time to Depletion of a Cell Phone Battery**

The objective of the model is to provide an estimation of time to full depletion of a Lithium-Ion cell phone battery while considering stochastic elements of human behavior and different use cases for the cell phone. Personalized recommendations on potential ways to increase battery life based on three separate use cases of the phone; Gamer, Socializer, and Worker are presented. The analysis focused on creating a reasonable way to model the randomness of human-phone interactions given user specified constraints thereby allowing the model to be adapted to represent any specific user's phone use habits.

The analysis was based on an implementation of the Coulomb-Counting method to estimate the state of charge (SOC) of a lithium-ion battery. The Coulomb-Counting method relates the instantaneous discharge current of a battery over time to a change in SOC. The model is applied to the Samsung Galaxy S3 phone (released 2012) because parameter values are well studied for this phone, however, the model can be applied to any set of specifications for more modern phones.

The focal point of this study is the unique way that user interactions with their phone are modeled. An algorithm which semi-randomly assigns usage times throughout the day for different particular user actions requiring power is presented. The algorithm provides a unique way to model the expected time until SOC reduces from 100% to 0% over the course of one day. Additionally, the model was run according to three different use case scenarios (Gamer, Socializer, and Worker) created to represent the different ways different people may use their phone including: duration of each use, intensity of each use, what kinds of apps are being run, and the kinds of background uses present.

Relevant recommendations are provided to the consumer tailored to each of the three design use cases. Since the model incorporates random elements a single model evaluation does not accurately represent the model findings. One thousand model runs were completed for each designed user (Gamer, Socializer, and Worker) in order to create a distribution of modeled time to complete discharge. The distributions show how the randomness of human interactions with their phone influence each user's daily time to complete discharge. Sensitivity of certain parameters were tested on each use case and new distributions were created and compared statistically to the base case. In this way the model can inform users about why and how frustrations with unpredictable battery life are related to more than just on screen usage.

The development of the SOC model predicting time to complete discharge was subject to the following major assumptions: Model parameters are used for the Galaxy S3 phone, open circuit voltage (OCV) of the Lithium-Ion battery is constant and not related to SOC, temperature does not impact the battery capacity under normal operating conditions, power usage of an application over time can be reasonably approximated by the nominal measured power usage, and use cases accurately model human behaviors.

Keywords: State of Charge, Lithium-Ion Battery, Stochastic, Coulomb-Counting, Smartphone

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1 Introduction

The following subsections will introduce the Background, Problem Statement, and Objective of Problem A of the 2026 CoMAP competition.

1.1 Background

Lithium-Ion batteries are now widely used since first becoming commercially available in the early 1990s. Lithium-Ion batteries can now be found in all kinds of electric vehicles, home devices, and nearly all mobile devices. Smartphones have become a cornerstone of modern life. According to data from the American Central Intelligence Agency there are approximately 9 billion active mobile-cellular device plans as of 2024.[1] People utilize their phones every day, for some it is an indispensable productivity tool for work or health, for others it is used for games and streaming to pass time. Despite their regular use, consumers are often confused and frustrated by the unpredictable behavior of their phone batteries, unsure of why some days or situations it depletes faster than others. Significant efforts have been made to model SOC of Lithium-Ion batteries in recent years because of their application to phone and electric vehicle battery management systems. Ultimately, battery depletion is complex and heavily dependent on the individual user settings, operating system, and app complexity/feature use.[2, 3]

The live performance of a battery cell depends on the current demand of the circuit, changing open circuit voltage, temperature, and the current physical characteristics of the battery influenced by its history such as the state of charge (SOC), state of health (SOH), and remaining useful life time (RUL) [4] [5]. Estimations of SOC are important for battery management systems inside mobile devices that make them more efficient as well as giving consumers an accurate understanding of the remaining level of charge and time-to-empty. Determining the SOC of a Lithium-Ion battery is a well studied subject and many different methods have been used for modeling SOC to provide accurate live estimations. Because of the importance of cell phones in daily life through the advent of health apps, work uses, social connection, and entertainment it is more important than ever that the consumer is able to understand and manage the SOC of their phone. A proper SOC estimate allows for a smooth user experience and optimization of the state of health (SOH) of a battery.

In its most basic form, state of charge is defined as the available battery capacity divided by the rated battery capacity as defined in Equation 1 [6]

$$SOC = \frac{C_{available}}{C_{rated}} \quad (1)$$

where,

$C_{available}$ is the remaining available charge in the battery and
 C_{rated} is the batteries rated charge.

1.2 Problem Statement

CoMAP will design a mathematical representation of the state of charge of a Lithium-Ion battery derived from a continuous time equation. The model must be able to use specifications of real phones

in order to estimate battery depletion under different use case scenarios. The team will develop realistic use case scenarios that model how people use their phones. The results of the model will be analyzed to answer the following research questions:

1. What is the SOC of the battery at a certain point in time given a specific use case?
2. How do different use cases effect the battery life?
3. How sensitive are model parameters to change?
4. What parameters most change time-to-empty estimations?

1.3 Objective

The objective of this report is to use a mathematical model to determine the state of charge (SOC) and predict the remaining time-to-empty of a smart phone battery. This will be done in the following steps.

1. Continuous-Time Model: Develop a model to represent SOC as a function of time. This model will incorporate factors such as screen usage, screen brightness, background task, and CPU load
2. Time-to-Empty predictions: Using the model, approximate time-to-empty using initial SOC and different usage scenarios
3. Use Cases: Develop use case scenarios that reflect the unpredictable human aspect of phone usage
4. Sensitivity Analysis: Determine how predictions vary by changing parameters and assumptions
5. Recommendations: Using the knowledge gained from the previous steps, develop practical recommendations for how to minimize battery drain throughout the day

2 Analysis of the Problem

The modeling of the SOC of a battery is a well studied field. There are four main kinds of Lithium-Ion battery models: [3, 7]

1. Filter methods (curve fitting)
2. Machine learning methods
3. Electrochemical models
4. Current integral models

The primary job of the CoMAP team is to determine which specific model is best suited for meeting the five objectives of the analysis listed in Section 1.3. A secondary consideration of the problem is how to interface the chosen model with the intrinsically stochastic nature of human behavior.

Filter and Machine Learning Methods:

Twelve different filter based methods are presented in [7]. Kalman Filters use regression and curve fitting on special model parameters in systems where there is uncertainty to account for random expected variations. Kalman filters are often applied to real time data to reduce error in live state predictions because of their efficiency.[8, 9] Live data with noise is not provided as part of this problem, so filter-based methods do not fit the needs of the project. Additionally, the estimation of model parameters using curve fitting and regression methods is explicitly against the rules of the CoMAP problem chosen. Machine learning and neural network models are also ruled out for this problem per the competition rules.

Electrochemical Models:

The third type of model considered was the electrochemical model which aims to describe battery charge/discharge by focusing on chemical reactions, anode/cathode properties. A difficulty related to this model discussed in source [10] is that modelers must have a deep understanding of the topic in order to develop models that accurately capture all of the physical processes influencing SOC. Our CoMAP team considered using the electrochemical model presented in [3], due to its relative simplicity as a first order system of ODEs yet robust description of charge discharge cycles. The model allowed for the tuning of discharge rates, efficiency adjustments, and accounted for changing open-circuit-voltage related to discharge. Ultimately, our CoMAP team did not use this model because the research did not describe the methods of obtaining or justifications for all parameter values used. The Lithium-Ion system modeled in the paper had a larger capacity and used a shorter time scale meaning new parameter values would be needed. Without proper description of parameter values and the potential need for curve fitting methods to estimate parameters, our CoMAP team could not justify the use of this novel model.

Current Integral Methods:

This leaves current integral models as our primary method for computing SOC. Current integral models, like the widely popular Coulomb Counting method, work by taking the integral of the instantaneous rate of current discharging from the battery at a given time t [6]. The model is simple but can be adapted multiple ways to decrease the error in measurements [11].

The relative ease of computation of the Coulomb Counting method allows for the modeling team to put additional efforts to expanding the robustness of the model. This is achieved by evaluation of results given the stochastic variation of power usage do to human nature.

Due to the dynamic range of applications that a mobile device can have depending on the user, multiple different use cases are needed to provide relevant feedback to the client on how to improve battery life depending on the scenario. Hence, the development of a method to generate relevant scenarios that target different components of a mobile device are needed for improved results.

3 Mathematical Model

The following section describes the formulation of the mathematical model and how it is run through the subsections Governing Equation, Designing Use Cases, Designing Users, Model Runs and Sensitivity,

and Model Assumptions.

3.1 Governing Equation

The relationship between SOC and demand of components causing battery depletion of the mobile device was developed using a Coulomb counting method derived in [12].

The Coulomb Counting method described in Equation 2 returns SOC as a percentage. It assumes a general battery discharge efficiency η . It should be noted that the Coulomb Counting method described in Equation 2 is an estimation and can be further improved in multiple ways as demonstrated by [7].

$$SOC(t) = SOC_0 - \frac{\eta}{C_n} \int_{t_0}^t I(t) dt * 100 \quad (2)$$

where,

SOC is the state of charge

SOC_0 is the state of charge at time zero

η is the efficiency of battery discharge to circuit

C_n is the batteries rated capacity and

$I(t)$ batteries instantaneous current discharge rate at time t .

$I(t)$ in reality is some continuous differentiable function representing the instantaneous current discharge from the battery. $I(t)$ in our model is defined as the summation of the instantaneous current discharge as calculated using Equation 4 and the parameters in Table 2.

The basic power equation is derived from Ohms law and states that power is the product of voltage and current as summarized in Equation 3.

$$P = V * I \quad (3)$$

where,

P is power in watts

V is voltage in volts and

I is current in amperes.

Our CoMAP team assumed that the open circuit voltage for the Galaxy S3 can be reasonably approximated as a constant value and does not change according to SOC [13]. This makes calculating the current for different use cases and smartphones a relatively simple calculation shown in Equation 4.

$$I(t) = \frac{\sum_{t=0}^n P_{usage}(t)}{V_b} \quad (4)$$

where,

$I(t)$ is batteries instantaneous discharge rate at time t in mA

t is time in min

$P_{usage}(t)$ is power usage for a particular smartphone operation as a function of time t in mW

V_b is battery voltage V

n is the number of smartphone operations in use at time t .

The summation of power demand at any time t can further be broken down into components, Φ and P as shown in Equation 5.

$$\sum_{t=0}^n P_{usage}(t) = \Phi_0 P_0(t_r) + \dots + \Phi_{n-1} P_{n-1}(t_r) + \Phi_n P_n(t_r) \quad (5)$$

where,

$P_{usage}(t)$ is power usage for a particular smartphone operation as a function of time t in mW

Φ is a modeled piecewise function describing the on-off state of a component

$P_n(t)$ is the function describing the power demand for a particular smartphone operation.

t_r is the local time variable describing length that a component is running

The function ϕ is critical to the success of the model because it represents the logic of whether or not the component is drawing power at time t . Importantly, $P_n(t)$ as power demand is not directly related to global time t but component runtime t_r . This t_r is then informed by the logic of the mask outlined in Section 3.2.

3.2 Designing Use Cases

The CoMAP team gave equal consideration to the importance of designing realistic, stochastic use cases that represent human usage of the phone. Due to the difficulty in obtaining reliable publicly available data for phone usage and the distribution throughout the day, a custom model was developed to create time of use functions for different phone applications and phone use habits.

A custom function was created in python to generate a controllable "mask" representing the piecewise wise function that describes the usage of each individual component of the cell phone. The mask function is used as a shortcut to generating many different continuous piecewise functions describing the state of usage of a smartphone operation. The function allows for many unique situations to be modeled by incorporating stochastic elements that change which operations are used at any given time every time the mask is created. Table 1 describes the variables used in the creation of the piece-wise mask functions ϕ_n .

Table 1: Summary of masking parameters: The following variables are used to create mask for the varying energy drain rates.

Variable Name	Description	Units
<i>Duration</i>	# of units of time to generate a mask for	min
<i>phone_time</i>	Amount of time units assigned as "active"	min
<i>block_range</i>	range (min,max) a component is on before turning off	min
<i>Exclusions</i>	Array of times where a particular P_n action cannot occur	unit-less

The step-by-step structure of the mask creation function is as follows:

1. create a 1D array of all zeros for every minute in desired *duration*
2. check for any excluded time locations
3. While *phone_time* > 0
 - Randomly choose a *block_length* within acceptable *block_range*. This represents the number of consecutive minutes a particular P_n is active.
4. Randomly select a time period equal to *block_length* within *duration* that is available for assignment
5. Update *Exclusions* array with the newly assigned active locations to avoid multiple assignments of the same time for the same component P_n
6. Repeat until all active *phone_time* time for a particular P_n is assigned a time period

The masking function is run for multiple different components, such that multiple unique mask arrays representing ϕ are returned. Each ϕ corresponds to a particular P_n indicating the rate of power usage for each. Hence, the two arrays can be combined to solve Equation 5 as shown in Equation 6 below.

$$\begin{array}{c} t_n \\ t_0 \\ \dots \\ t_{n-1} \\ t_n \end{array} \begin{array}{c} \phi_0 \quad \dots \quad \phi_{n-1} \quad \phi_n \\ \left[\begin{array}{ccccc} 0 & \dots & 1 & 1 \\ \dots & \dots & \dots & \dots \\ 1 & \dots & 0 & 1 \\ 1 & \dots & 1 & 1 \end{array} \right] \end{array} \times \begin{array}{c} P_n \\ \left[\begin{array}{c} P_0 \\ \dots \\ P_{n-1} \\ P_n \end{array} \right] \end{array} = \begin{array}{c} P_{usage}(t) \\ \left[\begin{array}{c} P_{usage}(t_0) \\ \dots \\ P_{usage}(t_{n-1}) \\ P_{usage}(t_n) \end{array} \right] \end{array} \quad (6)$$

where,

t_n is the time step

ϕ_n is the mask for indicating the state of a component at time t_n

P_n is an array of the nominal rate of power consumption by a component

$P_{usage}(t)$ is overall instantaneous power consumption at any given time step t_n

Once piecewise functions ϕ_n have been generated for each type of power usage (P_n) and the above matrix is solved for the given time step, $I(t)$ can be numerically solved for according to Equation 4. Each function ϕ_n indicating whether a component is on or off represents some set of individually continuous and differentiable piece-wise functions which are determined using the algorithm in Section 3.2.

3.3 Designing Users

Instead of making general recommendations on ways that users can improve battery lifespan, the CoMap team has designed several "Users". A user will help determine what parameters are important in different use cases. Without a designed user, the mask outlined in the previous section is meaningless and it is assumed that the system is fully activated and draining from $t = 0$, to $t = \infty$. Not having a user also makes recommendations for battery life trivial. Generally it is valid to recommend that high demand usage be reduced. To a user who plays a lot of games (often the highest demand on battery life), a suggestion that they should reduce game usage is not valuable. By developing a model that

more accurately represents that user, the CoMAP team can make recommendations that are practical.

To design a user, the following design considerations are important:

- How high demand are the users apps
- Duration of app usage
- Does the user use more foreground apps or background apps
- How frequently is the phone used
- What are the circumstances of phone usage
- What are the constant sources of battery drain

For this paper, the CoMAP team have designed several users that will be used in the model. The users have been dubbed the Gamer, the Socializer, and the Worker. These three users represent the high, medium, and low use cases [14].

The Gamer represents a high demand use case, where the majority of battery demand is through the use of high CPU/screen usage. The largest demand for CPU/screen usage is via gaming or watching videos on the phone. In this case we determined that a few long, high demand sessions throughout the day. The Gamer case uses a total of 3 hours of gaming and 2 hours of high quality video streaming over the course of a 14 hour day. See Table 3.

The Socializer is a medium activity user. What we notice in our reference is that while GPU/screen usage decreases, background processes stay relatively frequent. We decided the high frequency, short uses throughout the day in the form of phone calls, SMS interactions, and occasional social media use would best represent a medium demand use case. The Socializer case uses a total of 3 hours of low quality video (to simulate social media usage), 2 hours of phone calls, and 1 hour of SMS messaging in 1 minute intervals over the course of a 14 hour day. See Table 3.

The Worker is the low demand use case. The majority of demand in this use case come in the form of background processes. Spread throughout the day, background music apps, email, web usage, texting, and calling are the primary battery drains that are modeled. The Worker case uses a total of 4 hours of listening to music (downloaded music), 1 hour of emailing (on WiFi for base case), 1 hour of surfing the web (on WiFi for base case), 1 hour of SMS messaging, and 1 hour of phone calls over the course of a 14 hour day. See Table 3.

3.4 Model Runs & Sensitivity Analysis

The model was run 1000 times for each use case scenario designed (Gamer, Socializer, and Worker) to generate a distribution of data. Each model run returns the amount of time that it takes for the user to drain the battery from 100% SOC to 0% SOC. The result of all 1000 model runs are saved and plotted into histograms so that the distribution of data can be visualized. The model is run 1000 times because the piecewise functions ϕ_n are generated in a semi-random way limited by the input parameters of the

mask generation function hence a single evaluation of the model would not be a sufficient description. The randomness of the model intrinsically satisfies the condition of the sensitivity analysis of Problem A: evaluation of fluctuations in usage patterns.

The sensitivity of the model is further explored by altering certain parameters of the model for each user (Gamer, Socializer, and Worker) and completing 1000 additional simulations each time a parameter is changed. The new distribution of data is then compared to the original distribution using a two sided t-test over a 95% confidence interval to determine if changing a certain parameter or behavior produces statistically significant differences in the mean time until battery drain.

3.5 Model Assumptions

The simplifications used in our CoMAP teams math model were categorized into mathematical assumptions, or physical assumptions. These assumptions are defined in the following sections.

3.5.1 Mathematical

The following mathematical assumptions were made to the model:

- Assume SOC_0 is always 100%
- Assume that the minimum time of phone usage is one minute

3.5.2 Physical

The following physical assumptions were made to the model:

- Constant open circuit voltage OCV: Lithium-Ion batteries supply voltage to the circuit in the phone as a function of the current SOC [7]. Our model assumes a constant OCV
- Power demand by an app can be reasonably approximated by a constant rate
- Temperature does not effect battery discharge within normal operating conditions as per [15]
- No applications are running in the background
- The phone is always consuming an "idle" power
- The phone is always drawing power for WiFi connection (unless otherwise specified)
- Certain applications cannot "stack" power usage (e.g. no video streaming and gaming at the same time)
- The rated battery voltage, V_b , is a constant 3.7 Volts as per [13]

3.5.3 Use Case Design

Assumptions made for use case design are derived through logical reasoning where citations were not found. Cited assumptions and logic based assumptions are both noted.

- Only 14 hours a day are available for the allocation of phone time
 - Phones are often charged overnight, meaning that battery depletion cannot be modeled during this time
- Only one on-screen application runs at a time
 - Ex: video streaming and gaming cannot occur at the same time

4 Parameters and Initial Conditions

The following section describes the parameters and initial conditions used in both facets of the model and are described in the subsections SOC Model and Case Study.

4.1 SOC Model

In order to be able to solve Equation 2, first the parameters representing the power usage rate of a component. The parameters needed for use found for use in the SOC model are listed in Table 2. η was calculated by taking the inverse of the measured battery discharge efficiency [16].

4.2 Use Case Model

For each of the three use cases different durations and ranges were selected to be randomly distributed throughout the 14 hour day. The duration dictates the total time the smartphone spent performing one operation throughout the course of the day. The range provides the minimum and maximum amount of time a smartphone operation can be performed in any one stretch. The amount of time the smartphone spent performing an operation was randomly chosen within its respective range. The duration and ranges for each operation are summarized in Table 3. It should be noted that idle power was active when the smartphone was performing no other tasks.

5 Numerical Method

The following section provides reasoning for the choice of using Euler's method (Equation 7) to model the given scenario. In order to solve Equation 2, a numerical integration method is required to estimate the value of the integral over time. As mentioned in Section 3.5 the rate that different phone applications demand power can be reasonably approximated by a constant, zero order equation. Because a zero order equation is being used and the slope is constant at zero, the ordinary differential equation describing the rate of change in the power demand is trivial and 0 for all t_n . As a result more powerful and computationally intense numerical methods such as higher order RK methods are not necessary to solve this problem because of the characteristic high stability [18]. Since only the constant rate of power demand per unit time active is needed to approximate overall change in SOC,

Table 2: Summary of Power Ratings for a Samsung Galaxy S3: The following power ratings were found experimentally. [16, 17, 14]

Specification	Description	Value	Units
C_n	Rated Battery Capacity	3900	Milliampere Hours
η	Measured Battery Efficiency	120	Percent
Idle	Idle phone w/ screen on	805	Milli-Watt
Screen	Isolated Screen	360	Milli-Watt
WiFi	Isolated WiFi	50	Milli-Watt
Video High	High quality video stream	3599	Milli-Watt
Video Low	Low quality video stream	2655	Milli-Watt
3D Game	Need for Speed	2425	Milli-Watt
Call	Audio Call	864	Milli-Watt
SMS	Texting w/ 3G connection	1140	Milli-Watt
Web WiFi	Using Web w/ WiFi connection	1275	Milli-Watt
Email WiFi	Using Email w/ WiFi connection	1264	Milli-Watt
Web 3G	Using Web w/ 3G connection	1479	Milli-Watt
Email 3G	Using Email w/ 3G connection	1543	Milli-Watt
Audio	Isolated Audio	226	Milli-Watt
Background	Nominal Background App Usage	150	Milli-Watt

a less computationally intense method such as Euler's is acceptable and allows for faster simulation run time. If $P_n(t_n)$ were modeled with a higher order equation a different numerical method may need to be explored as the rate of power consumption would be changing with time, forming a first order differential equation.

$$y_{n+1} = y_n + h \frac{\delta i}{\delta t} \quad (7)$$

where,

y_{n+1} is the value of function at next time step

y_n is the value of the function at the current time step and

$\frac{\delta i}{\delta t}$ is power draw per unit time.

6 Validating the Model

In order to ensure an accurate and robust model, validations of the SOC Equations, Numerical Method, Use Case Development, and Parameters are needed.

Table 3: Summary of Duration and Ranges for Random Smartphone Operations. Operation is the smartphone operation, duration is the total time the phone is performing the operation throughout the day, and range is the span in which the operation is active. The span of the range is chosen randomly for each seed. All times are presented in minutes.

Gamer			Socializer			Worker		
Operation	Duration	Range	Operation	Duration	Range	Operation	Duration	Range
3D Game	180	30-120	LQ Video	180	1-60	Music	240	1-60
HQ Video	120	10-60	Call	120	1-120	Email	60	1-15
Idle	540	NA	SMS	60	1-2	Web	60	1-30
-	-	-	Idle	480	NA	SMS	60	1-2
-	-	-	-	-	-	Call	60	1-60
-	-	-	-	-	-	Idle	360	NA

6.1 SOC Equations

The CoMAP team grappled with determining if the model presented was in fact a continuous-time-model or discrete and assume that the judges may have similar confusion, however it is maintained that the model does represent a continuous time model for the reasons described in this subsection.

The creation of randomized "Use Cases" or unique user scenarios for use in this model can be thought of as a shortcut to the process of meticulously creating and defining many small, continuous, piecewise functions to fit the descriptions of the Gamer, Socializer, and Worker use cases. It happens that in the case of this model the functions used are zero order, or constant. A zero order function satisfies the definition of a continuous function given in [19]. The model may be more clearly thought of as continuous if instead of a zero order function used to model power usage of a component over time $P_n(t)$ the power usage was a higher order function such as a cubic, or sin wave function. The ODE for our case as mentioned in Section 5 is trivial as it is equal to zero so it is not explicitly used in solving the problem. Due to these reasons, the group maintains that both methods are valid.

6.2 Numerical Method

The Euler's method program used to integrate Equation 7 [18] was verified against a problem with a known solution to test for validity. The problem chosen was a linear function shown in Equation 8 below. A time step of 1 unit was assumed.

$$y = 5x + 3 \text{ for } 0 \leq x \leq 10 \quad (8)$$

The program verified the known solution of 280 units.

6.3 Use Case Development

Each user's load (Gamer, Socializer, and Worker) on the battery is informed by [14] and reasonable logical conclusions about which functions of the phone cannot happen at the same time.

6.4 Parameters

We used the Samsung S3 parameters due to the publicly available data from [16]. Our CoMAP teams model can be easily altered to represent newer smart phone models as more data becomes available to the public.

7 Results

This section will provide the results generated by our CoMAP team's mathematical model.

7.1 Use Case Results

The math model was applied to three different use cases, the Gamer, the Socializer, and the Worker. The Gamer represents a high use case, the Socializer a medium use case, and the Worker a low use case. The masking functions ϕ_n were used to describe the distribution of power demand by components over a day. The summed power demand of all the components at an instant t was calculated. Then for any time t in the day, the total SOC reduction was calculated by integrating $P_{usage}(t)$ from t_o to a particular t_n where t is in minutes. Thus allowing for minute by minute SOC estimations used to show the SOC of the smart phone battery over the course of a day.

7.1.1 Gamer Use Case Results

The baseline case for the Gamer utilized three energy consumption rates. Idle with screen off, 3D game (Need for Speed), and streaming a high quality video. See Table 2 for measured power ratings. The baseline case did not account for background app usage. Figure 1 shows the estimated time-to-empty for each minute the phone was on for a single seed.

1000 seeds were run using the Gamer baseline case. Figure 2 shows the distribution of the time it takes to completely drain the smartphone battery using the Gamer use case. The average time to complete drain is 295 minutes.

7.1.2 Socializer Use Case Results

The baseline case for the Socializer utilized four energy consumption rates. Idle with screen off, SMS messaging with a 3G connection, calling audio only, and streaming a low quality video. See Table 2 for measured power ratings. The baseline case did not account for background app usage. 1000 seeds were run using the Socializer baseline case. Figure 3 shows the distribution of the time it takes to completely drain the smartphone battery using the Socializer use case. The average time to complete drain is 365 minutes.

7.1.3 Worker Use Case Results

The baseline case for the Worker utilized six energy consumption rates. Idle with screen off, SMS messaging with a 3G connection, calling audio only, emailing with a WiFi connection, surfing the web with a WiFi connection, and listing to music (assuming it's downloaded). See Table 2 for measured

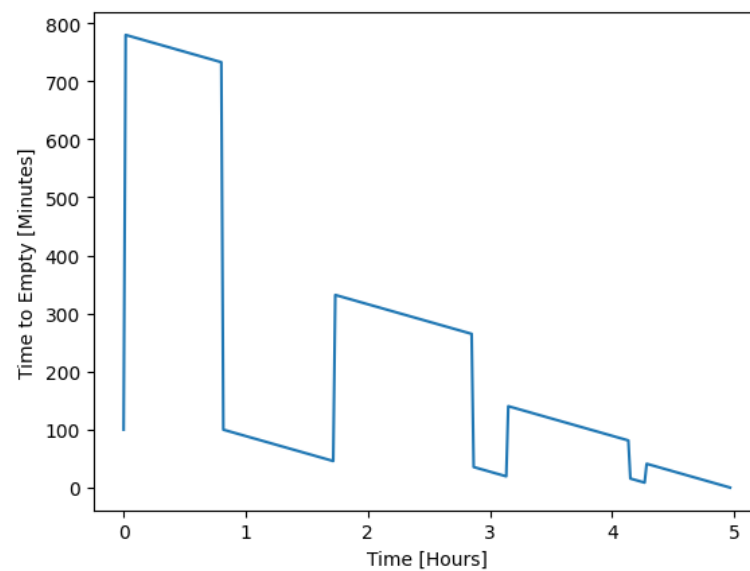


Figure 1: time-to-empty vs Time for the Gamer case. The x-axis represents the time since the phone has been completely charged and the y-axis represents the estimated time until empty based off of the current energy drain rate. This figure was generated using a stochastic method to allocate altering power usages over time. It represents one out of a thousand sample seeds. The battery lasted approximately 5 hours without charging. It should be noted that this particular TTE plot shares a seed with Figures 5 and 6.

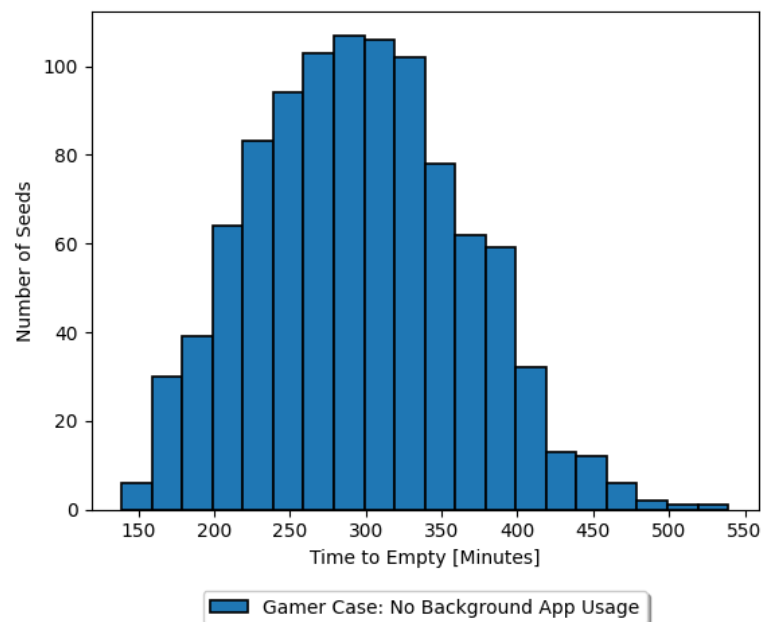


Figure 2: Gamer time-to-empty vs Number of Seeds. The x-axis represents the time it took for the battery to fully drain in minutes and the y-axis number of seeds in each bin. The bin width is 20 minutes. This figure was generated using a stochastic method to allocate altering power usages over time for 1000 seeds. These seeds do not account for background app usage. The average time to complete drain is 295 minutes.

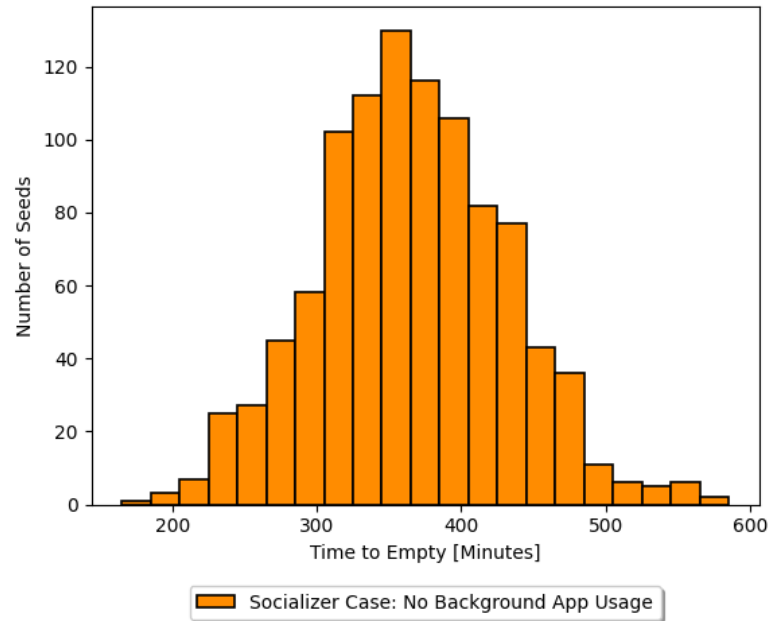


Figure 3: Socializer time-to-empty vs Number of Seeds. The x-axis represents the time it took for the battery to fully drain in minutes and the y-axis number of seeds in each bin. The bin width is 20 minutes. This figure was generated using a stochastic method to allocate altering power usages over time for 1000 seeds. These seeds do not account for background app usage. The average time to complete drain is 365 minutes.

power ratings. The baseline case did not account for background app usage. 1000 seeds were run using the Worker baseline case. Figure 4 shows the distribution of the time it takes to completely drain the smartphone battery using the Worker use case. The average time to complete drain is 449 minutes.

7.1.4 Comparison between Use Case Results

For the Gamer case, three energy consumption rates were utilized. Idle with screen off, 3D game (Need for Speed), and streaming a high quality video. For the Socializer case, four energy consumption rates were used. Idle with screen off, SMS messaging with a 3G connection, calling (audio only), and streaming a low quality video. For the Worker case, six energy consumption rates were used. Idle with screen off, SMS messaging with a 3G connection, calling (audio only), emailing with a WiFi connection, surfing the web with a WiFi connection, and listing to music (assuming it's downloaded). These measured power ratings can be found in Table 2. The power demand over time for a random seed for each of the three use cases were run and plotted against one another in Figure 5. These cases were ran with no background app usage in an effort to simplify the model. Using the same random seed, the SOC for each of the three cases were plotted against one another in Figure 6. These seeds did not account for background app usage. These figures are not representative of the entire distribution but rather a single seed out of 1000.

The baseline cases were rerun with a nominal background app usage of 150 mW. Figure 7 shows the power demand over time for all three use cases utilizing a nominal background app usage of a single random seed. Figure 8 shows the state of charge as a function of time for the same random seed as shown in Figure 7.

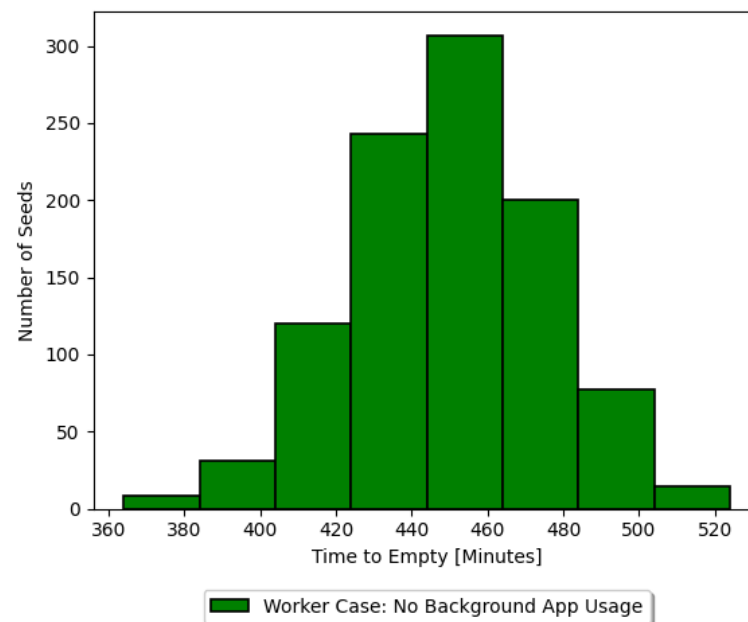


Figure 4: Worker time-to-empty vs Number of Seeds. The x-axis represents the time it took for the battery to fully drain in minutes and the y-axis number of seeds in each bin. The bin width is 20 minutes. This figure was generated using a stochastic method to allocate altering power usages over time for 1000 seeds. These seeds do not account for background app usage. The average time to complete drain is 449 minutes.

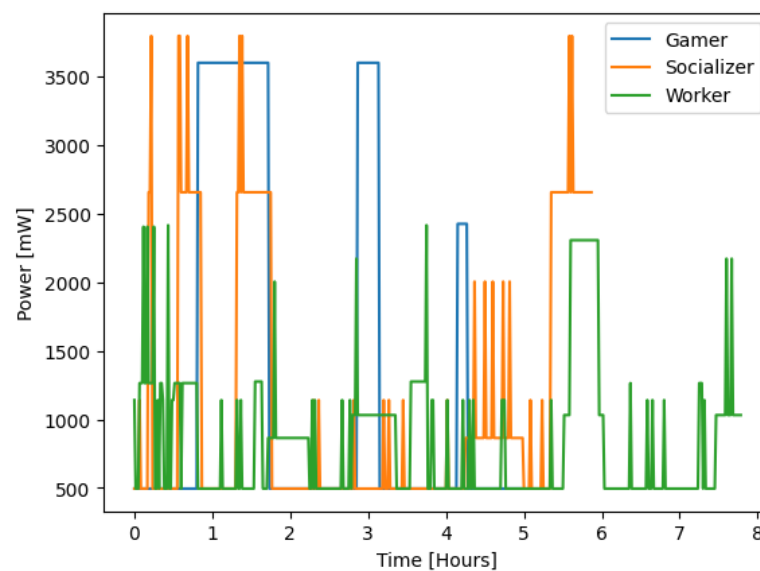


Figure 5: Power vs Time for all three use cases. The x-axis represents the time since the phone has been completely charged and the y-axis represents the power rating used at that particular minute. This figure was generated using a stochastic method to allocate altering power usages over time. For the gamer, socializer, and worker use cases the battery lasted approximately 5, 6, and 8 hours without charging respectively. Each line represents one out of a thousand sample seeds. This particular seed disregarded background app usage.

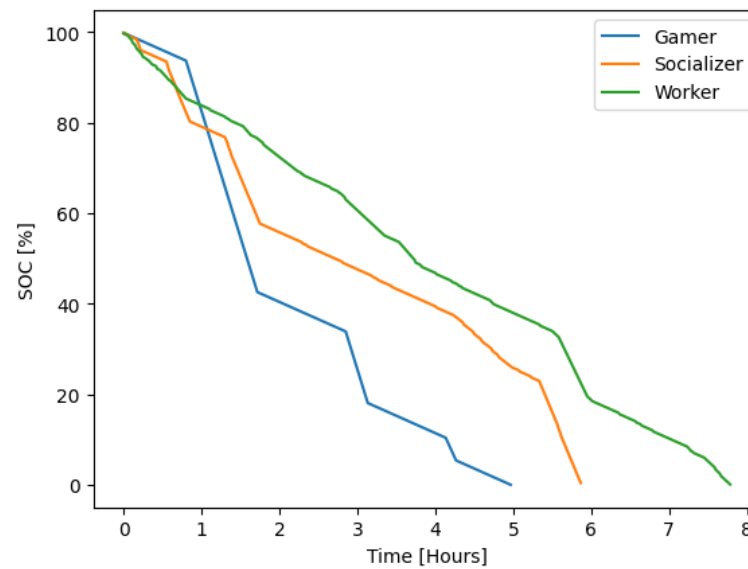


Figure 6: SOC vs Time for all three use cases. The x-axis represents the time since the phone has been completely charged and the y-axis represents remaining charge in the battery at time t . This figure was generated using a stochastic method to allocate altering power usages over time. For the gamer, socializer, and worker use cases the battery lasted approximately 5, 6, and 8 hours without charging respectively. Each line represents one out of a thousand sample seeds. This particular seed disregarded background app usage.

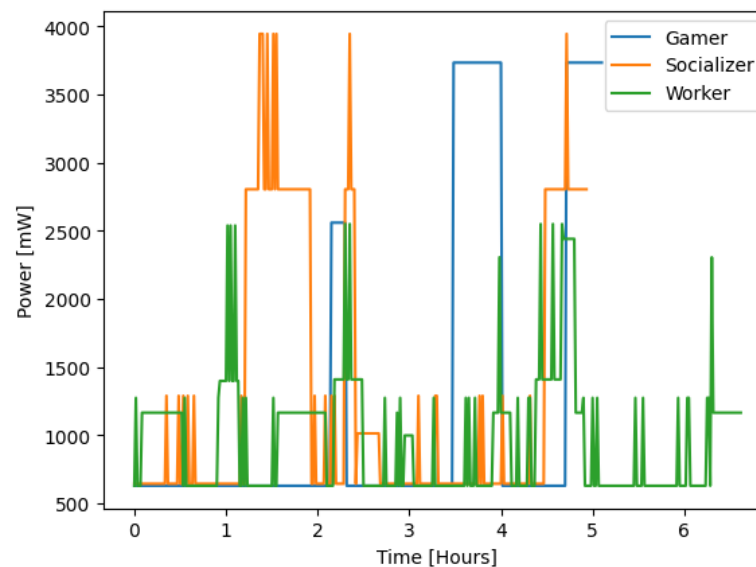


Figure 7: Power vs Time for all three use cases. The x-axis represents the time since the phone has been completely charged and the y-axis represents the power rating used at that particular minute. This figure was generated using a stochastic method to allocate altering power usages over time. For the gamer, socializer, and worker use cases the battery lasted approximately 5, 5, and 7 hours without charging respectively. Each line represents one out of a thousand sample seeds. This particular seed **accounts** for background app usage.

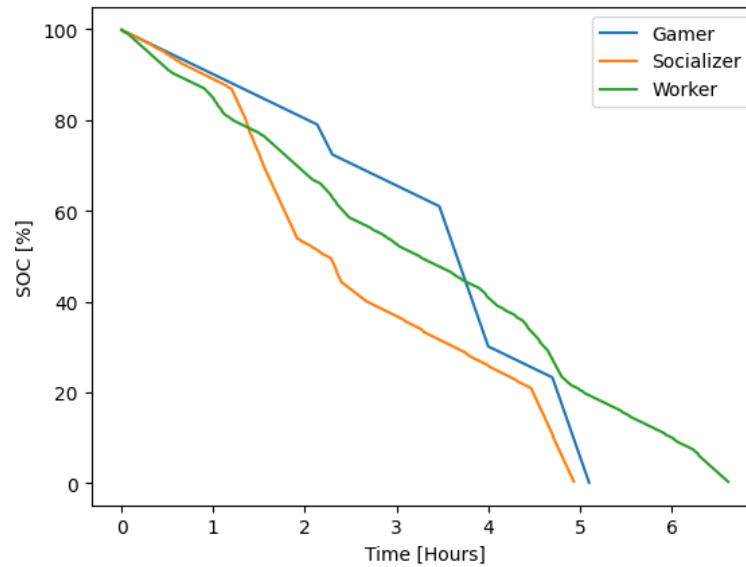


Figure 8: SOC vs Time for all three use cases. The x-axis represents the time since the phone has been completely charged and the y-axis represents remaining charge in the battery at time t . This figure was generated using a stochastic method to allocate altering power usages over time. For the gamer, socializer, and worker use cases the battery lasted approximately 5, 5, and 7 hours without charging respectively. Each line represents one out of a thousand sample seeds. This particular seed **accounts** for background app usage.

1000 random seeds for each of the three new cases were run and compared. For the Gamer, Socializer, and Worker the time it took for the battery to drain on average was 295, 365, and 449 minutes respectively. Figure 9 shows the random distribution of the time-to-empty for each of the three use cases. The bin width for each case is 20 minutes. The worker case has a much narrower distribution due to the smaller block ranges selected for each smart phone operation.

7.1.5 Sensitivity Analysis

A sensitivity analysis was performed on the parameters that can be controlled by the user of a smart-phone. These included, whether or not background apps were turned on, a comparison between WiFi and 3G connection, as well as whether screen brightness was at its maximum or minimum setting. Every sensitivity was run with a sample size of 1000. The averages for each case, including baseline are summarized in Table 4.

Each run was then compared to the baseline using a p-test to determine if changing a particular variable produces results that are statistically significant from the baseline. Statistical significance indicates that the variable does have an impact on SOC and thus is sensitive to change.

8 Discussion

The aim of the Discussion section is to discuss the results presented in Results section.

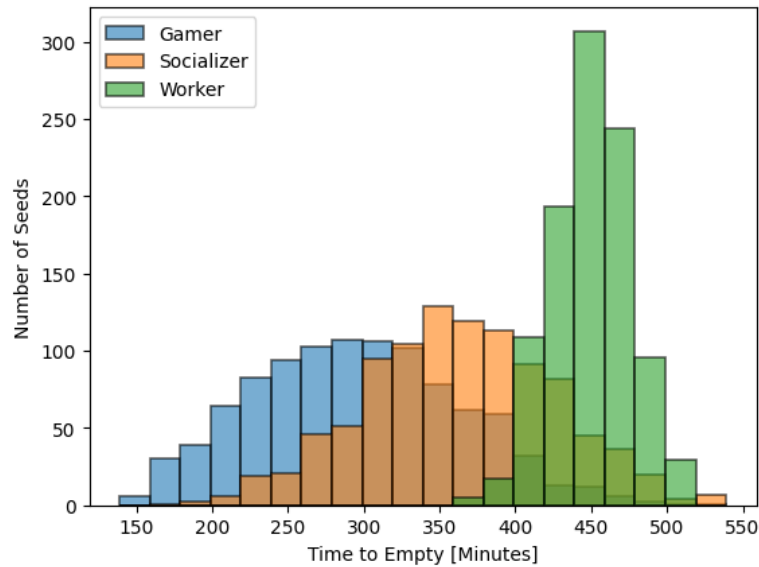


Figure 9: time-to-empty vs Number of Seeds for all three use cases. The x-axis represents the time it took for the battery to fully drain in minutes and the y-axis number of seeds in each bin. The bin width for each case is 20 minutes. This figure was generated using a stochastic method to allocate altering power usages over time for 1000 seeds for each of the three cases. These seeds do not account for background app usage. The average time to complete drain for the Gamer, Socializer, and Worker is 295, 365, and 449 minutes respectively.

8.1 Base Model

The base model was run 1000 times for each use case (Gamer, Socializer, and Worker) in order to find the distribution of the time to depletion of the battery. The resulting histograms for all three cases are overlaid and shown in Figure 9. All three distributions shown are normally distributed. The fact that all distributions are normal helps to validate the modeling approach taken to model human-phone interactions and is supported by real world observations. Most of the time a persons phone will fully deplete the battery at the mean depletion time, however depending on random factors such as the duration of each interaction with the phone, the stacking of app usage (email and music simultaneously), and the distribution of interactions throughout the day, different results will ultimately be achieved every day. The model is further validated by the fact that the heavy use case of Gamer generally depletes battery faster then the Socializer who has short but frequent on-screen interactions and both deplete faster than the Worker who has more time of use but less on-screen time.

8.2 Sensitivity Analysis

The specific value of the change in the average time-to-empty for the sensitivity analysis are summarized in Table 4. The results of the sensitivity analysis show that in all cases reducing screen brightness increases the time-to-empty of the battery when examined on a 95% confidence interval with $\alpha = 0.05$. The p-values are listed in Table 5 and are all less than $\frac{\alpha}{2}$. This evaluates the extreme case of maximum brightness versus minimum brightness. Teh extreme case is still useful because it shows that adjusting screen brightness is something that can cause significant changes of time-to-empty. Therefore, even small changes can have a measurable impact. Similar conclusions are found for background app usage for all use cases.

Table 4: Summary of Smartphone User Options: The three main use cases were run with varying smartphone settings changed. The average time-to-empty for each case is listed below. Each case has a sample size of 1000. BG stands for background. 3G App & No WiFi means that the smartphone's WiFi was turned off and all app power ratings were calculated using their 3G rates. All times are presented in minutes.

Gamer		Socializer		Worker	
Parameter	AVG TTE	Parameter	AVG TTE	Parameter	AVG TTE
Baseline	295.427	Baseline	364.687	Baseline	449.154
BG App Usage	270.345	BG App Usage	320.612	BG App Usage	389.487
Low Brightness	306.411	Low Brightness	371.95	Low Brightness	468.153
No WiFi	302.925	3G App & No WiFi	338.206	Stream Music	431.903

Table 5: P-values of tested variables: The P-values of from each model run. The P-values indicate statistical significance if less than 0.025.

Gamer		Socializer		Worker	
Parameter	p-value	Parameter	p-value	Parameter	p-value
Baseline	-	Baseline	-	Baseline	-
BG App Usage	1.2054e-17	BG App Usage	9.6638e-58	BG App Usage	≈ 0
Low Brightness	0.0004	Low Brightness	0.0128	Low Brightness	2.62e-53
No WiFi	0.0148	3G App & No WiFi	1.0652e-21	Stream Music	7.43e-116

Each of the three use cases was also adjusted in a unique way pertinent to their specifications. (1) The Gamer was made to no longer consume power for WiFi in the background, indicating a user choice to switch WiFi off when not needed. (2) The Worker was made to use 3G data for emailing and web browsing instead of WiFi meant to represent the impacts of working from a non-central location. (3) The Socializer was made to connect to 3G data for the entire day instead of connecting to WiFi for the entire day meant to represent times when the individual is out and about. All three changes caused a significant change in the distributions when compared to the base case of the model runs. The Gamer had a longer time-to-empty with WiFi off, the Worker and Socializer both had a shorter time-to-empty when substituting 3G data instead of WiFi in their individual cases. The specific value of the change in mean time-to-empty for the sensitivity analysis are summarized in Table 4.

8.3 Strengths and Weaknesses

Strengths:

Representing the overall current leaving the battery as a summation of the total demand of the system allowed the use of a mask to connect the continuous nature of power draw in a phone to the discontinuous nature of phone use by humans. Using the mask created a "plug and play" model, where the CoMAP team could explore different interactions between the user and between different components of the phone. By validating a linear approximation of power draw for the different components modeled, the relationship between the mask, the components, and the use over the day became an parameter driven discretizations that allowed for the use of thousands of model runs. These model runs could then be compared to each other utilizing statistical analysis to identify what parameters have the largest effect on SOC.

Weaknesses:

A main weakness of the Coulomb-Counting SOC model is the integration method used over time. In this case Euler's was used. $I(t)$, the instantaneous current demand of phone operation was modeled as a zero order function. The model in its current state will require some changes in order to accurately characterize non-linear power demand. In non-linear scenarios Euler's method causes drift due to error over time and must be re calibrated more often compared to methods like RK45 [20]. The age of the data used to validate and create the model means that the recommendations provided are not necessarily true for current phones. The use of the Galaxy S3 was decided because of the availability of data related to the phone. The phone itself was released in 2012, not a reflection of the capability of current phones. The model currently assumes a linear relationship between the power draw of each individual component and time. This means there are some nuances in power demand increases over time that are being lost in our model. Time limitations prevented the study of a more complex model that includes more dynamic component behavior, which would include developing a method to keep track of run time. When approximating the integral of current in the coulomb counting method as a summation, an amount of error is introduced that was not accounted for in this paper.

8.4 Recommendations

Recommendations for how to improve battery life were primarily informed by the sensitivity analysis. Overall, the biggest recommendation for improving battery life is to reduce the quantity of background applications running at any time. Reducing the number of background applications increased the average lifespan by 11%, representing an increase of about an extra hour of battery life for each user. Other parameters had a smaller impact, but when maximizing battery life the following recommendations can be made. Users utilizing high demand applications over long periods of time similar to the Gamer can improve battery life by turning off WiFi when not in use, and reducing screen brightness as is reasonable. Users who have phone usages similar to the Worker or the Socializer can improve battery life by lowering screen brightness, and utilizing WiFi instead of Data when possible. For the Socializer and the Worker, cellular data proved to be a high battery drain over the day. While turning off data completely isn't always feasible, minimizing the amount of time a phone is connected to or searching for cellular data connections will improve battery time-to-empty.

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9 AI Use Report

The following tasks were performed by an AI model:

- Our CoMAP teams report was written in Overleaf, a cloud based LaTeX editing software. Overleaf features an AI model that suggests edits to users, fixing things like grammar and sentence structure. We used this feature very minimally.
- Chat GPT was used to find relevant sources. Our team provided the model with sources we found and asked it to provide us with other similar sources. All used sources found with the model have been cited and are from verified publishers.
- Chat GPT was used to help format Tables 3, 4, and 5. The values presented in Tables 3, 4, and 5 were generated by our CoMAP team and are not the work of an AI model.